

**BABU BANARASI DAS UNIVERSITY
LUCKNOW**



**SUBJECT NAME : - Apply end to end security of cloud
Application**

SUBJECT CODE :-BCACSN14201

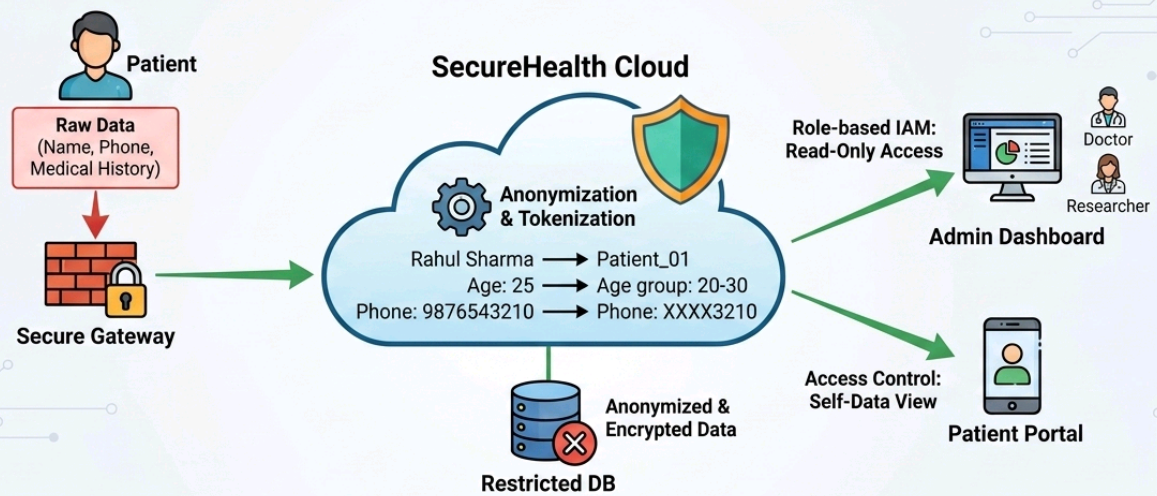
**PROJECT NAME : - SecureHealth: Privacy-Preserving
Cloud Portal**

**SUBMITTED TO -
MR. SAUVIK DHALI SIR**

**SUBMITTED BY -
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Professional Infographic

SecureHealth: Privacy-Preserving Healthcare Cloud Portal.



- Docker Containerized Deployment
- AWS Cloud Architecture
- IAM-driven Access Policies
- VPC Security Groups for Isolated Database

Clone from GitHub

💡 Project Idea and Features

📁 Code available in /src

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Sub- Apply end-to-end Security & Cloud

IBM Project 4th Sem.

Project Name- "Secure Health Cloud Portal".

Object:- Implementing End-to-End Security in Cloud
- Native Healthcare Applications.

① Introduction:- Healthcare organizations store highly sensitive patient data, including personal details, medical history and financial records. Storing this data in the cloud makes it a prime target for cyber-attacks.

This project demonstrates How to protect such data using advanced security techniques like. Anonymization, Tokenization and VPC Isolation.

② Key Security Features:

① Data Anonymization: Automatically modifies personal information so that individuals cannot be identified (e.g. changing a name to "Patient").

② Data Masking: Hide sensitive fields on the UI. (e.g. Phone: 8081XXXXX11).

③ Containers Security:- Packaging the application using "Docker" for consistent and secure deployment.

③ Technical Stack:-

- ① Cloud Provider:- AWS (VPC, IAM, EC2, S3).
- ② Frontend: React.js.
- ③ Back-end: Node.js/Express.
- ④ Database: Mongo DB Atlas (Encrypted at Rest).
- ⑤ Dev Ops: Docker & Kubernetes.
- ⑥ Security Logic: Crypto JS for encryption and JWT for session management

④ Feature Enhancements:

- ① AI-Driven Intrusion Detection:- Implementing an IDS to monitor Network traffic for suspicious patterns
- ② Zero-Trust Architecture:- Verifying every request even if it comes from inside the network.
- ③ Blockchain Integration:- To maintain immutable logs of who accessed which patient record.